

$$\int_2^5 x^5 \ln(x) dx$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$dv = x^5 dx$$

$$v = \frac{1}{6} x^6$$

$$\ln x \cdot \frac{1}{6} x^6 - \int \frac{1}{6} \cdot \frac{x^6}{x} dx$$

$$\ln x \cdot \frac{1}{6} x^6 - \left[\frac{1}{6} \int x^5 dx \right]$$

$$\ln x \cdot \frac{1}{6} x^6 - \frac{1}{6} \left[\frac{1}{6} x^6 \right]$$

$$\ln x \cdot \frac{x^6}{6} - \frac{x^6}{36} \Big|_2^5$$

$$\left((\ln(2)) \cdot \frac{2^6}{6} - \frac{2^6}{36} \right) - \left((\ln(2)) \cdot \frac{1}{6} - \frac{1}{36} \right)$$

$$\left((\ln(2)) \cdot \frac{64}{6} - \frac{64}{36} \right) - \left((\ln(2)) \cdot \frac{1}{6} - \frac{1}{36} \right)$$

$$\left(\frac{64}{6} \ln(2) - \frac{64}{36} \right) - \left(\frac{1}{6} \ln(2) - \frac{1}{36} \right)$$

how does this become $-\frac{7}{4}$?

$$\ln(2) = 0$$

$$\frac{64}{6} \cdot \frac{6}{6} \ln(2) - \frac{7}{4} = \frac{64}{36} \ln(2^6) - \frac{7}{4}$$

property of logs

$$9 \ln(6) = \ln(6^9)$$

$$\rightarrow \frac{64}{6} \cdot \frac{1}{6} \ln(2^6) - \frac{7}{4}$$

$$= \frac{26}{9} \ln(64) - \frac{7}{4}$$

$$= \frac{16 \ln(64)}{9} - \frac{7}{4}$$